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Efficiency in constrained continuous location ¹

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Abstract

We present a geometrical characterization of the efficient, weakly efficient and strictly efficient points for multiobjective location problems in presence of convex constraints and when distances are measured by an arbitrary norm. These results, established for a compact set of demand points, generalize similar characterizations previously obtained for unconstrained problems. They are used to show that, in planar problems, the set of constrained weakly efficient points always coincides with the closest projection of the set of unconstrained weakly efficient points onto the feasible set. This projection property which are known previously only for strictly convex norms, allows to easily construct all the weakly efficient points and provides a useful localization property for efficient and strictly efficient points. © 1998 Elsevier Science B.V.

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1. Introduction

Efficiency is a classical solution concept to deal with multicriteria optimization problems. In the framework of location analysis this concept has been extensively studied (see e.g. [1–11]). Out of its own interest to determine the best possible locations with respect to some set A of demand points, strict efficiency has been used to develop localization results for the solution to the Weber problem or its multifacility version (see e.g. [12–14]). In

addition, as clearly shown in [15,16], weak efficiency plays a crucial role in competitive location to find Condorcet points or Simpson points. More precisely, evaluating the best possible market share of a firm in the case of entry of a second firm in the market requires the complete determination of the sets of weakly efficient points relative to some subsets of the demand points. Thus developing characterizations of efficiency allowing to easily construct the whole efficient set is of prime importance. In this paper we address the convex constraint case and give such characterizations for strictly efficient, efficient and weakly efficient points. We also study if efficient points have the *projection property* or in other words if the set of

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constrained efficient points coincides with the closest projection of the set of unconstrained efficient points onto the feasible set.

Throughout the paper X is a real normed space. The norm is denoted by γ and the notation $\langle \cdot, \cdot \rangle$ is used for the duality bracket between X and its dual X^* . Though this assumption is not essential, at least in the first part of the paper, the reader may assume that X is finite dimensional or even that $X = \mathbb{R}^n$ in which case X^* will be identified to X and $\langle \cdot, \cdot \rangle$ is the usual scalar product.

We will recall now some classical results of convex analysis. We refer the reader to [17] for these notions. The set $B := \{x \in X; \gamma(x) \leq 1\}$, whose subset of extreme points will be denoted by $\text{Ext}(B)$, is called the unit ball of γ . If B is a polytope, γ is said to be polyhedral and if B is strictly convex, i.e. if the unit sphere $S := \{x \in X; \gamma(x) = 1\}$ contains no (nontrivial) segment, γ is called strictly convex. Given a direction $\delta \in X$, γ is said to possess the property B_δ if for each x such that $\gamma(x - \delta) \leq \gamma(x)$, there exists a strictly positive real λ and a neighborhood W of x such that $\gamma(y - \lambda\delta) \leq \gamma(y)$ for every $y \in W$. If the property B_δ is satisfied for each $\delta \neq 0$, γ will be called B-regular (regular in the sense of Brown [18]). For example, strictly convex and polyhedral norms in any dimension are B-regular as well as arbitrary norms in dimension 2 [7].

For a subset $\Omega \subset X$ we will denote respectively by $\text{int}(\Omega)$, $\bar{\Omega}$, $\text{co}(\Omega)$, $\overline{\text{co}}(\Omega)$ and by $\Omega^0 := \{p \in X^*; \langle p, y \rangle \leq 1, \forall y \in \Omega\}$, the interior, the closure, the convex hull, the closed convex hull and the polar set of Ω . We recall that the directional derivative of a convex function $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ defined by

$$f'(x; \delta) := \lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{1}{t} (f(x + t\delta) - f(x))$$

is the support function of its subdifferential denoted by $\partial f(x)$, which means that we have the useful equality

$$f'(x; \delta) = \max_{p \in \partial f(x)} \langle p, \delta \rangle,$$

where

$$\partial f(x) := \{p \in X^*; \forall y \in X, f(y) \geq f(x) + \langle p, y - x \rangle\}.$$

We recall that the subdifferential of a norm is given by

$$\partial \gamma(x) = \{p \in B^0; \langle p, x \rangle = \gamma(x)\}.$$

Let $\Omega \subset X$ be a nonempty closed convex set. If $x \in \Omega$, we denote by $N_\Omega(x)$ the normal cone to Ω at x , i.e. the convex cone defined by

$$N_\Omega(x) := \{p \in X^*; \forall y \in \Omega, \langle p, y - x \rangle \leq 0\}$$

or equivalently by $N_\Omega(x) = \partial \Psi_\Omega(x)$ where the indicator function Ψ_Ω is defined by $\Psi_\Omega(x) = 0$ if $x \in \Omega$ and $\Psi_\Omega(x) = +\infty$ if $x \notin \Omega$. We call set of feasible directions at $x \in \Omega$ the set

$$F_\Omega(x) := \{\delta \in X; \delta \neq 0, \exists \lambda > 0, x + \lambda\delta \in \Omega\}.$$

If $x \in \Omega$ and if δ is a feasible direction at x , the point $x + \lambda\delta$ remains in Ω for any $\lambda > 0$ sufficiently small. We denote by $\lambda_\delta := \sup\{\lambda \geq 0; x + \lambda\delta \in \Omega\}$ the largest λ having this property. We also call set of tangent directions at $x \in \Omega$ the set

$$T_\Omega(x) := \left\{ \delta \in X; \delta \neq 0, \exists (\delta_n), \exists (\lambda_n) > 0 \right. \\ \left. \text{such that } x + \lambda_n \delta_n \in \Omega, \delta = \lim_{n \rightarrow \infty} \delta_n \right\}.$$

Note that $T_\Omega(x) = \overline{F_\Omega(x)}$ and that $T_\Omega(x)$ can be viewed as the polar set of $N_\Omega(x)$, i.e. $T_\Omega(x) = (N_\Omega(x))^0$.

For $a \in X$, we call projection of a onto Ω (with respect to γ) the set

$$\text{Proj}_\Omega^\gamma(a) := \text{argmin}\{\gamma(a - x), x \in \Omega\}$$

and if $A \subset X$ we put $\text{Proj}_\Omega^\gamma(A) := \bigcup_{a \in A} \text{Proj}_\Omega^\gamma(a)$.

Further, we will need the following notation. For $\delta \in X$, $\delta \neq 0$, we denote by H_δ^+ and H_δ^- the open half spaces of X^* defined by

$$H_\delta^+ := \{p \in X^*; \langle p, \delta \rangle > 0\}, \\ H_\delta^- := \{p \in X^*; \langle p, \delta \rangle < 0\}.$$

2. Definitions and first properties

Definition 2.1. Let $A \subset X$ and let $\Omega \subset X$ (a set of constraints), $A, \Omega \neq \emptyset$.

(i) A point $x \in \Omega$ is said to be strictly efficient with respect to A and Ω if

$\forall y \in \Omega, y \neq x, \exists a \in A, \gamma(a-x) < \gamma(a-y).$

(ii) A point $x \in \Omega$ is said to be efficient with respect to A and Ω if

$$\forall y \in \Omega, y \neq x, \begin{cases} \exists a \in A, \gamma(a-x) < \gamma(a-y) \\ \text{or} \\ \forall a \in A, \gamma(a-x) \leq \gamma(a-y). \end{cases}$$

(iii) A point $x \in \Omega$ is said to be weakly efficient with respect to A and Ω if

$$\forall y \in \Omega, y \neq x, \exists a \in A, \gamma(a-x) \leq \gamma(a-y).$$

The set of strictly efficient, efficient and weakly efficient points with respect to A and Ω will be, respectively, denoted by $\text{SE}(A, \Omega)$, $\text{E}(A, \Omega)$ and $\text{WE}(A, \Omega)$. In the unconstrained case, i.e. when $\Omega = X$, these sets are more simply denoted by $\text{SE}(A)$, $\text{E}(A)$ and $\text{WE}(A)$.

Below we give without proof a list of elementary (but useful) properties.

Proposition 2.1. *Let $A, B \subset X$, $A, B \neq \emptyset$, and let Ω and Ω' be two nonempty subsets of X . Then the following properties hold.*

- (1) *If Ω is closed, then $\text{E}(A, \Omega) \neq \emptyset$, $\text{WE}(A, \Omega) \neq \emptyset$ but the nonemptiness of $\text{SE}(A, \Omega)$ is not guaranteed.*
- (2) *$\text{SE}(A) \cap \Omega \subset \text{SE}(A, \Omega)$, $\text{E}(A) \cap \Omega \subset \text{E}(A, \Omega)$ and $\text{WE}(A) \cap \Omega \subset \text{WE}(A, \Omega)$.*
- (3) *$A \cap \Omega \subset \text{SE}(A, \Omega) \subset \text{E}(A, \Omega) \subset \text{WE}(A, \Omega)$.*
- (4) *$\text{SE}(A-x, \Omega) = \text{SE}(A, \Omega+x) - x$, $\text{E}(A-x, \Omega) = \text{E}(A, \Omega+x) - x$ and $\text{WE}(A-x, \Omega) = \text{WE}(A, \Omega+x) - x$.*
- (5) *If $A \subset B$, then $\text{SE}(A, \Omega) \subset \text{SE}(B, \Omega)$, $\text{WE}(A, \Omega) \subset \text{WE}(B, \Omega)$ but such a property does not necessarily hold for efficient points.*
- (6) *If $\Omega \subset \Omega'$, we have $\text{SE}(A, \Omega') \subset \text{SE}(A, \Omega)$, $\text{E}(A, \Omega') \subset \text{E}(A, \Omega)$ and $\text{WE}(A, \Omega') \subset \text{WE}(A, \Omega)$.*
- (7) *If A is bounded, then $\text{SE}(A, \Omega)$, $\text{E}(A, \Omega)$ and $\text{WE}(A, \Omega)$ are bounded.*
- (8) *$\text{SE}(A) \cap \text{int}(\Omega) = \text{SE}(A, \Omega) \cap \text{int}(\Omega)$, $\text{E}(A) \cap \text{int}(\Omega) = \text{E}(A, \Omega) \cap \text{int}(\Omega)$ and $\text{WE}(A) \cap \text{int}(\Omega) = \text{WE}(A, \Omega) \cap \text{int}(\Omega)$.*
- (9) *$\text{SE}(A, \Omega) = \text{SE}(\bar{A}, \Omega)$, $\text{E}(A, \Omega) = \text{E}(\bar{A}, \Omega)$ but such a property does not necessarily hold for weakly efficient points.*

Example. Let $X = \mathbb{R}^2$, γ be the ℓ_1 norm, and consider the points $a_1 = (0, 0)$, $a_2 = (2, 3)$, $a_3 = (3, 2)$, $a_4 = (3, 4)$, $b_1 = (1, 1)$, $b_2 = (4, 4)$, $c_1 = (2, 2)$, $c_2 = (3, 3)$ and $c_3 = (2, 4)$.

- If $A = \{a_1, a_2, a_3, a_4\}$ and $\Omega = [b_1, b_2]$ then $\text{SE}(A, \Omega) = \text{E}(A, \Omega) = [b_1, c_2]$ and $\text{WE}(A, \Omega) = [b_1, b_2]$.
- If $A = \{a_2\}$ and $\Omega = [b_1, b_2]$ then $\text{SE}(A, \Omega) = \emptyset$, the sets of efficient and weakly efficient points coincide and we have $\text{E}(A, \Omega) = \text{WE}(A, \Omega) = [c_1, c_2]$.
- If $\Omega = \text{co}\{c_1, a_3, c_2\}$, $x = (\frac{5}{2}, \frac{9}{4})$, then with $A = \{a_2, a_3\}$ we have $\text{E}(A) = \text{co}\{c_1, a_2, c_2, a_3\}$, and $x \in \Omega = \text{E}(A) \cap \Omega \subset \text{E}(A, \Omega)$ while with $A' = \{a_1, a_2, a_3\}$, $x \notin \text{E}(A', \Omega)$.
- If $\Omega = [c_1, c_2]$, $A =]a_2, a_3[$, then $c_1 \in \text{WE}(\bar{A}, \Omega)$ but $c_1 \notin \text{WE}(A, \Omega)$. By symmetry the same holds for c_2 .

A first characterization of weak efficiency, given by Carrizosa and Plastria in [4] is the following.

Proposition 2.2. [4] *Assume that X is finite dimensional and let $A \subset X$ be a nonempty compact set, $\Omega \subset X$ be a nonempty closed convex set, $x \in \Omega$. Then*

$$x \in \text{WE}(A, \Omega) \iff 0 \in \text{co} \left(\bigcup_{a \in A} \partial \gamma(x-a) \right) + N_{\Omega}(x).$$

3. Efficient points in terms of Q_{δ} sets

As shown in [7] a collection of cones can be associated with the norm γ . These cones are closely related to the geometry of the unit ball B and play an important role in the geometrical description of the sets of efficient points developed in this section.

Definition 3.1. For any $\delta \in X$, $\delta \neq 0$, according to [7] we consider the two complementary sets

$$Q_{\delta} := \{z \in X; \forall \lambda > 0, \gamma(z - \lambda\delta) > \gamma(z)\},$$

$$P_{\delta} := \{z \in X; \exists \lambda > 0, \gamma(z - \lambda\delta) \leq \gamma(z)\}.$$

It is easily shown that Q_{δ} and P_{δ} are cones. More precisely, Q_{δ} is the cone generated by the points x of the unit sphere S such that we leave the unit ball B when moving from x in the direction

$-\delta$. Topological and geometrical properties of the Q_δ sets have been studied in detail in [7]. Let us recall some of these results which will be useful later.

The Q_δ sets are closed as soon as the norm is B-regular, e.g. in dimension two. However some Q_δ sets associated with exotic norms may be not closed as illustrated by an example in [7]. We always have $\text{int}(Q_{-\delta}) \subset P_\delta$ (the inclusion may be strict) and $Q_\delta = \overline{P_{-\delta}}$ if the unit sphere S contains no segment parallel to δ (e.g. when γ is a strictly convex norm). If X is a Hilbert space, it is easy to check that $Q_\delta = \overline{P_{-\delta}} = \overline{H_\delta^-}$. The following proposition gathers several characterizations of the interior of a Q_δ set which can be found in [5,6] or in [7].

Proposition 3.1. *For each $\delta \in X$, $\delta \neq 0$, we have the following equivalences:*

- (i) $x \in \text{int}(Q_{-\delta})$,
- (ii) $x \notin \overline{P_{-\delta}}$,
- (iii) $\exists \lambda > 0$, $\gamma(x - \lambda\delta) < \gamma(x)$,
- (iv) $\partial\gamma(x) \subset H_\delta^+$.

Proof. (i) $\Leftrightarrow$ (ii): The proof is obvious and can easily be obtained by complementarity. (i) $\Leftrightarrow$ (iii): The proof can be found in [7]. (iii) $\Rightarrow$ (iv): Let x such that $\gamma(x - \lambda\delta) < \gamma(x)$ for some $\lambda > 0$ and let p be an arbitrary subgradient in $\partial\gamma(x)$. Then

$$\gamma(x - \lambda\delta) \geq \gamma(x) + \langle p, -\lambda\delta \rangle$$

and we obtain the strict inequality $\langle p, \delta \rangle > 0$ which means that $p \in H_\delta^+$.

(iv) $\Rightarrow$ (iii): If $\partial\gamma(x) \subset H_\delta^+$ we have

$$\gamma'(x; -\delta) = \max_{p \in \partial\gamma(x)} \langle p, -\delta \rangle < 0$$

and therefore

$$\exists \lambda > 0, \gamma(x - \lambda\delta) < \gamma(x). \quad \square$$

Concerning the geometrical properties, the two dimensional case is of particular interest because the Q_δ sets are convex, closed, and can be easily described. Any given direction $\delta \neq 0$ generates two parallel supporting lines to B , respectively, denoted by D and D' (by symmetry we obviously have $D' = -D$). If the face of B defined by $F := D \cap B$ (resp. $F' := D' \cap B$) contains a single point $u \in \text{Ext}(B)$ (resp. by symmetry $-u \in \text{Ext}(B)$),

and if we denote by $J(u)$ the closed half plane limited by the line passing through u and $-u$ and which does not contain δ , then $Q_\delta = J(u)$ and $\overline{P_\delta} = -J(u)$. Note that in the Hilbert case we have $Q_\delta = J(u) = \overline{H_\delta^-}$.

If F is a segment $[u, v]$ with $u, v \in \text{Ext}(B)$ and $v = u + \lambda\delta$ for some $\lambda > 0$, we have $F' = [-u, -v]$ and Q_δ is the acute convex cone generated by u and $-v$. Equivalently, we have $Q_\delta = J(u) \cap J(v)$ and $\overline{P_{-\delta}} = J(u) \cup J(v)$ where $J(u)$ and $J(v)$ are the closed half planes, respectively, induced by u and v and defined as previously.

Note that if B was not symmetric or equivalently if the property $\gamma(x) = \gamma(-x)$ was not satisfied (in which case γ is called a gauge and not a norm), some Q_δ sets may be nonconvex. As example, in the plane the gauge γ whose unit ball is $B = \text{co}\{e_1, e_2, e_3, e_4\}$ with $e_1 = (2, 2)$, $e_2 = (-4, 4)$, $e_3 = (-3, -1)$ and $e_4 = (2, -1)$ admits four nonconvex Q_δ sets, e.g. the one generated by the direction $\delta = (0, 1)$.

Now, let us characterize weak efficiency.

Proposition 3.2. *Let $A \subset X$ be a nonempty compact set, Ω be a nonempty closed convex set and let $x \in \Omega$. Then*

$$x \in \text{WE}(A, \Omega) \iff \forall \delta \in F_\Omega(x), A \cap (x + \overline{P_{-\delta}}) \neq \emptyset.$$

Proof. Suppose that $x \notin \text{WE}(A, \Omega)$. Then by definition of $\text{WE}(A, \Omega)$ we have

$$\exists y \in \Omega \text{ such that } \forall a \in A, \gamma(a - y) < \gamma(a - x).$$

Taking $\delta = y - x$ we get $\delta \in F_\Omega(x)$ and

$$\forall a \in A, \gamma(a - x - \delta) < \gamma(a - x).$$

Hence, by Proposition 3.1, replacing x by $a - x$ for each $a \in A$, we deduce that $A \subset x + \text{int}(Q_{-\delta})$ or equivalently $A \cap (x + \overline{P_{-\delta}}) = \emptyset$.

Now suppose that there exists $\delta \in F_\Omega(x)$ such that $A \cap (x + \overline{P_{-\delta}}) = \emptyset$. Thus, by Proposition 3.1 we have

$$\forall a \in A, \exists \lambda_a > 0, \gamma(a - x - \lambda_a\delta) < \gamma(a - x).$$

For each $a \in A$, the function $y \rightarrow \gamma(a - y)$ is continuous. Thus there exists a neighborhood $V(a)$ of a such that $\gamma(a' - x - \lambda_a\delta) < \gamma(a' - x)$ for

any $a' \in V(a)$. The set A being compact, there exists a finite set of demand points $A' \subset A$ such that A is covered by $\bigcup_{a' \in A'} V(a')$. Let us choose any λ such that $0 < \lambda \leq \min\{\min\{\lambda_{a'}, a' \in A'\}, \lambda_\delta\}$. The norm γ being convex, we obtain

$$\forall a \in A, \gamma(a - x - \lambda\delta) < \gamma(a - x).$$

Observing that $y = x + \lambda\delta \in \Omega$, we conclude that $x \notin \text{WE}(A, \Omega)$ and the proof is complete. $\square$

Corollary 3.1. *If $A \subset X$ is a nonempty compact set, $\Omega \subset X$ is a nonempty closed convex set and $x \in \Omega$, then*

$$x \in \text{WE}(A, \Omega) \iff \forall \delta \in T_\Omega(x), A \cap (x + \overline{P_{-\delta}}) \neq \emptyset.$$

Proof. As $F_\Omega(x) \subset T_\Omega(x)$, we only have show that if A meets $x + \overline{P_{-\delta}}$ for each feasible direction, the same property holds for any tangent direction. So, assume that

$$\forall \delta \in F_\Omega(x), A \cap (x + \overline{P_{-\delta}}) \neq \emptyset$$

and consider $\delta \in T_\Omega(x)$. By definition of $T_\Omega(x)$ there exists a sequence (δ_n) of feasible directions (i.e. $\delta_n \in F_\Omega(x)$) converging to δ . Thus, for each $n \in \mathbb{N}$, $A \cap (x + \overline{P_{-\delta_n}}) \neq \emptyset$, and, by contraposition of Proposition 3.1, we can find some $a_n \in A$ satisfying the inequality

$$\forall \lambda > 0, \gamma(a_n - x) \leq \gamma(a_n - x - \lambda\delta_n).$$

The set A being compact, we can assume without loss of generality that the sequence (a_n) converges to some $\bar{a} \in A$ and using continuity arguments we get

$$\forall \lambda > 0, \gamma(\bar{a} - x) \leq \gamma(\bar{a} - x - \lambda\delta).$$

Henceforth $\bar{a} \in x + \overline{P_{-\delta}}$ and then $A \cap x + \overline{P_{-\delta}} \neq \emptyset$. $\square$

In dimension 2 and in the unconstrained case, Proposition 3.2 has several interesting consequences already proved in [7] and given in Corollary 3.2. Note in particular that $\text{WE}(A)$ can be regarded as an intersection of half planes containing A , is convex, contains $\text{co}(A)$ and thus enjoys nice hull properties providing a simple means to construct it geometrically.

Corollary 3.2. [7] *If X is two dimensional and $A \subset X$ is a nonempty compact set, then*

$$\begin{aligned} \text{WE}(A) &= \{x \in X, \forall \delta \neq 0, A \cap (x + \overline{P_{-\delta}}) \neq \emptyset\} \\ &= \{x \in X, \forall u \in \text{Ext}(B), A \cap (x + J(u)) \neq \emptyset\} \\ &= \bigcap_{\substack{A \subset x + J(u) \\ x \in X, u \in \text{Ext}(B)}} (x + J(u)) \end{aligned}$$

and $\text{WE}(\text{WE}(A)) = \text{WE}(A)$.

Further useful characterizations of weak efficiency can be given in the unconstrained case.

Proposition 3.3. *Let $A \subset X$ be a nonempty compact set. Then the following statements are equivalent:*

- (i) $x \notin \text{WE}(A)$,
- (ii) $\exists \delta \neq 0, A \subset x + \text{int}(Q_{-\delta})$,
- (iii) $\exists \delta \neq 0, \bigcup_{a \in A} \partial\gamma(a - x) \subset H_\delta^+$,
- (iv) $0 \notin \text{co}\left(\bigcup_{a \in A} \partial\gamma(a - x)\right)$,
- (v) $0 \notin \bigcup_{a \in \text{WE}(A)} \partial\gamma(a - x)$.

If X is two dimensional one may add the additional conditions:

- (vi) $\exists \delta \neq 0, \text{WE}(A) \subset x + \text{int}(Q_{-\delta})$,
- (vii) $\exists \delta \neq 0, \bigcup_{a \in \text{WE}(A)} \partial\gamma(a - x) \subset H_\delta^+$,
- (viii) $0 \notin \text{co}\left(\bigcup_{a \in \text{WE}(A)} \partial\gamma(a - x)\right)$.

Proof. (i) $\Leftrightarrow$ (ii): The result directly follows from the characterization of $\text{WE}(A)$ in terms of Q_δ sets (see Proposition 3.2).

(ii) $\Leftrightarrow$ (iii): This equivalence is a direct consequence of the characterization of $\text{int}(Q_\delta)$ (see Proposition 3.1)

(iii) $\Leftrightarrow$ (iv): Assume that $0 \notin \text{co}(\bigcup_{a \in A} \partial\gamma(a - x))$. As this set is closed and convex, a classical separation theorem [17] shows that there exists some $\delta \neq 0$ such that $\bigcup_{a \in A} \partial\gamma(a - x) \subset H_\delta^+$.

Conversely if $\bigcup_{a \in A} \partial\gamma(a - x) \subset H_\delta^+$ for some $\delta \neq 0$ then its convex hull is also contained in H_δ^+ and thus $0 \notin \text{co}(\bigcup_{a \in A} \partial\gamma(a - x))$.

(i) $\Leftrightarrow$ (v): Observe that $0 \in \partial\gamma(x)$ if and only if $x = 0$ and the result follows. Note that the equivalence between (i) and (iv) is also a direct conse-

quence of Proposition 2.2 applied in the unconstrained case ($\Omega = X$).

Now if the dimension of X is two we have the equality $\text{WE}(\text{WE}(A)) = \text{WE}(A)$ (see Corollary 3.2). This closure property allows to substitute A by $\text{WE}(A)$ in the previous equivalences (ii), (iii) and (iv) and this completes the proof. $\square$

Strictly efficient and efficient points can be characterized in a similar way provided that the norm is assumed to be $\mathbf{B}$ -regular. Observe that this assumption is satisfied for the most commonly used norms in the context of location. In particular, all norms (gauges) are $\mathbf{B}$ -regular in dimension 2 as well as strictly convex and polyhedral norms in any dimension as noted in Section 1.

Proposition 3.4. *Let $A \subset X$ be a nonempty compact set, $\Omega \subset X$ be a nonempty closed convex set and $x \in \Omega$. If γ is $\mathbf{B}$ -regular, then*

- (i) $x \in \text{E}(A, \Omega) \iff \forall \delta \in F_\Omega(x),$
 $A \cap (x + Q_\delta) \neq \emptyset$ or $A \cap (x + \text{int}(Q_{-\delta})) = \emptyset.$
- (ii) $x \in \text{SE}(A, \Omega) \iff \forall \delta \in F_\Omega(x),$
 $A \cap (x + \Omega_\delta) \neq \emptyset.$

Proof. The justifications of (i) and (ii) being very similar, we only give the proof of (i) which is slightly more complicated. Suppose that $x \notin \text{E}(A, \Omega)$. Then by definition of $\text{E}(A, \Omega)$

$$\exists y \in \Omega, y \neq x$$

$$\text{such that } \begin{cases} \forall a \in A, & \gamma(a - y) \leq \gamma(a - x) \\ \text{and} \\ \exists a \in A, & \gamma(a - y) < \gamma(a - x). \end{cases}$$

Setting $\delta = y - x$ and $\lambda = 1$ one gets $\delta \in F_\Omega(x)$ and

$$\forall a \in A, \quad \gamma(a - x - \lambda\delta) \leq \gamma(a - x),$$

$$\exists a \in A, \quad \gamma(a - x - \lambda\delta) < \gamma(a - x),$$

which exactly means that $A \cap (x + Q_\delta) = \emptyset$ and $A \cap (x + \text{int}(Q_{-\delta})) \neq \emptyset$.

Now suppose that $A \cap (x + Q_\delta) = \emptyset$ and $A \cap (x + \text{int}(Q_{-\delta})) \neq \emptyset$ for some direction $\delta \in F_\Omega(x)$. Then

$$\forall a \in A, \quad \exists \lambda_a > 0, \quad \gamma(a - x - \lambda_a\delta) \leq \gamma(a - x),$$

$$\exists \bar{a} \in A, \quad \exists \bar{\lambda}, \quad \gamma(\bar{a} - x - \bar{\lambda}\delta) < \gamma(\bar{a} - x).$$

Since γ is assumed to be $\mathbf{B}$ -regular, for each $a \in A$, there exists a neighborhood $V(a)$ of a such that

$$\exists \mu_a > 0, \quad \forall a' \in V(a), \quad \gamma(a' - x - \mu_a\lambda_a\delta) \leq \gamma(a' - x).$$

Setting $\mu'_a = \mu_a\lambda_a$ and using the convexity of γ , we deduce

$$\forall \lambda \in [0, \mu'_a], \quad \forall a' \in V(a), \quad \gamma(a' - x - \lambda\delta) \leq \gamma(a' - x),$$

$$\forall \lambda \in]0, \bar{\lambda}], \quad \gamma(\bar{a} - x - \lambda\delta) < \gamma(\bar{a} - x).$$

The set A being compact, there exists a finite set of demand points $A' \subset A$ such that A is covered by $\bigcup_{a' \in A'} V(a')$. Choosing any λ such that $0 < \lambda \leq \min\{\min\{\mu'_{a'}, a' \in A'\}, \bar{\lambda}, \lambda_\delta\}$ we see that

$$\forall a \in A, \quad \gamma(a - x - \lambda\delta) \leq \gamma(a - x)$$

$$\exists a \in A, \quad \gamma(\bar{a} - x - \lambda\delta) < \gamma(\bar{a} - x).$$

By construction $y = x + \lambda\delta \in \Omega$, thus we deduce that $x \notin \text{E}(A, \Omega)$ and the proof is complete. $\square$

Remark 3.1. In Proposition 3.4, we cannot replace $F_\Omega(x)$ by $T_\Omega(x)$. To illustrate this claim consider the space $X = \mathbb{R}^2$, the set A containing the single demand point $a = (0, 1)$ and the set constraints $\Omega = D_1 \cap D_2$ where D_1 (resp. D_2) is the disk centered at $c_1 = (1, 1)$ (resp. $c_2 = (1, -1)$) and with radius $r = \sqrt{2}$. If γ is the ℓ_1 norm we have $T_\Omega(0) = \{(x_1, x_2); x_2 \leq x_1, x_2 \geq -x_1\}$ and $F_\Omega(0) = \text{int}(T_\Omega(0)) = \{(x_1, x_2); x_2 < x_1, x_2 > -x_1\}$. The origin is a strictly efficient point but all the directions $\delta \in T_\Omega(0)$ generate only three different Q_δ sets, only two of which intersect A .

In the strictly convex case it is well known that $\text{SE}(A) = \text{E}(A) = \text{WE}(A)$ (see e.g. [7]). In the presence of (convex) constraints this result remains valid.

Proposition 3.5. *Let $A \subset X$ be a nonempty compact set, $\Omega \subset X$ be a nonempty closed convex set and assume that γ is strictly convex. Then*

$$\text{SE}(A, \Omega) = \text{E}(A, \Omega) = \text{WE}(A, \Omega).$$

Proof. From Propositions 3.2 and 3.4, the result is a consequence of the fact that, if the norm is strictly convex, we have $Q_\delta = \overline{P_{-\delta}}$ for each direction δ . Note that a direct proof similar to that of Proposition 1.3 in [7] could also be given. $\square$

4. Constrained efficient points as closest projections

4.1. Characterizations of closest projection

Let us start with several useful characterizations of the projection of a point onto a closed convex set.

Proposition 4.1. *Let $\Omega \subset X$ be a nonempty closed convex set, $a \in X$ and $x \in \Omega$. Then the following are equivalent:*

- (i) $x \in \text{Proj}_\Omega^\gamma(a)$,
- (ii) $-N_\Omega(x) \cap \partial\gamma(x-a) \neq \emptyset$,
- (iii) $a-x \in \bigcap_{\delta \in F_\Omega(x)} \overline{P_{-\delta}}$.

Proof. (i) $\Leftrightarrow$ (ii): The result is a direct consequence of the equivalences

$$\begin{aligned} x \in \text{Proj}_\Omega^\gamma(a) &\Leftrightarrow x \in \text{argmin} \{ \gamma(y-a) \\ &\quad + \Psi_\Omega(y), y \in X \} \\ &\Leftrightarrow 0 \in \partial(\gamma(\cdot-a) + \psi(\cdot))(x) \\ &\Leftrightarrow 0 \in \partial\gamma(x-a) + N_\Omega(x) \\ &\Leftrightarrow -N_\Omega(x) \cap \partial\gamma(x-a) \neq \emptyset. \end{aligned}$$

(i) $\Leftrightarrow$ (iii): The result is a direct consequence of the equivalences

$$\begin{aligned} x \in \text{Proj}_\Omega^\gamma(a) &\Leftrightarrow \forall y \in \Omega, \quad \gamma(x-a) \leq \gamma(y-a) \\ &\Leftrightarrow \forall \delta \in F_\Omega(x), \forall \lambda \in [0, \lambda_\delta], \\ &\quad \gamma(a-x) \leq \gamma(a-x-\lambda\delta) \\ &\Leftrightarrow \forall \delta \in F_\Omega(x), \forall \lambda > 0, \\ &\quad \gamma(a-x) \leq \gamma(a-x-\lambda\delta) \\ &\Leftrightarrow \forall \delta \in F_\Omega(x), \quad a-x \in \overline{P_{-\delta}} \\ &\Leftrightarrow a-x \in \bigcap_{\delta \in F_\Omega(x)} \overline{P_{-\delta}}. \end{aligned}$$

Note that the second assertion implies the third one in the equivalences above due to the convexity of the norm. $\square$

Remark 4.1. In the Hilbert case, it is easy to verify that the set

$$N_\Omega^\gamma(x) := \bigcap_{\delta \in F_\Omega(x)} \overline{P_{-\delta}}$$

coincides with $N_\Omega(x)$ and thus, condition (iii) of Proposition 4.1 reduces to the classical characterization of the projection,

$$x = \text{Proj}_\Omega(a) \Leftrightarrow a-x \in N_\Omega(x).$$

In the general case, $N_\Omega^\gamma(x)$, which is a cone depending on the geometrical structure of the feasible set Ω and on the geometry of the norm γ , can be viewed as a generalization of the concept of normal cone. This cone is always convex in dimension two when the norm is strictly convex (since each $\overline{P_{-\delta}}$ is a half space) but may be nonconvex in the other cases, in particular when the norm is polyhedral.

4.2. General projection properties

The next result which shows that in the Hilbert case efficient (hence strictly and weakly efficient) points have the *projection property* has already been established in finite dimension in [2]. Here we give a new and completely different proof valid in the infinite dimensional case.

Theorem 4.1. *Let X be a Hilbert space, $A \subset X$ be a nonempty compact set and $\Omega \subset X$ be a nonempty closed convex set. Then we have*

$$\text{WE}(A, \Omega) = \text{Proj}_\Omega^\gamma(\text{WE}(A)) = \text{Proj}_\Omega^\gamma(\overline{\text{co}}(A)).$$

Proof. In Hilbert spaces, it is well known that $\text{WE}(A) = \overline{\text{co}}(A)$. Thus the second equality holds. Let us now prove the first equality and let $x \in \text{Proj}_\Omega^\gamma(\text{WE}(A))$. Since γ is strictly convex there exists $\bar{a} \in \text{WE}(A)$ such that $\{x\} = \text{Proj}_\Omega^\gamma(\bar{a})$ and we have

$$\forall \delta \in F_\Omega(x), \forall \lambda \in]0, \lambda_\delta], \quad \gamma(\bar{a}-x) < \gamma(\bar{a}-x-\lambda\delta).$$

By convexity of γ we get for every $\delta \in F_\Omega(x)$,

$$\forall \lambda > 0, \quad \gamma(\bar{a}-x) < \gamma(\bar{a}-x-\lambda\delta),$$

and then $\bar{a} \in x + Q_\delta$. Since $\bar{a} \in \text{WE}(A)$, Proposition 3.2 applied with $\Omega = X$ shows that there exists $a \in A$ such that $a \in \bar{a} + \overline{P_{-\delta}}$. Observing that $Q_\delta = \overline{H_\delta} = \overline{P_{-\delta}}$ in any Hilbert space, we get $a \in (x + \overline{P_{-\delta}}) + \overline{P_{-\delta}} = x + \overline{P_{-\delta}}$ and then $A \cap (x +$

$\overline{P_{-\delta}} \neq \emptyset$ for every $\delta \in F_{\Omega}(x)$, which means again by Proposition 3.2 that $x \in \text{WE}(A, \Omega)$.

Conversely, assume that $x \notin \text{Proj}_{\Omega}^{\gamma}(\text{WE}(A))$. From condition (iii) of Proposition 4.1 and according to Remark 4.1 we see that $(\text{WE}(A) - x) \cap N_{\Omega}(x) = \emptyset$. The cone $N_{\Omega}(x)$ being closed and convex while the set $\text{WE}(A) - x$ is convex and compact (since $\text{WE}(A) = \overline{\text{co}}(A)$), a classical separation theorem [17] shows that

$$\exists \delta \in X, \delta \neq 0, \exists b \in \mathbb{R}$$

$$\text{such that } \begin{cases} \langle \delta, y \rangle < b, & \forall y \in N_{\Omega}(x), \\ \langle \delta, y \rangle > b, & \forall y \in \text{WE}(A) - x. \end{cases}$$

Since the linear form $y \rightarrow \langle y, \delta \rangle$ is bounded from above on the cone $N_{\Omega}(x)$, it must reach its maximum on $N_{\Omega}(x)$ at the origin. Thus we obtain

$$\begin{cases} \langle \delta, y \rangle \leq 0, & \forall y \in N_{\Omega}(x), \\ \langle \delta, y \rangle > 0, & \forall y \in \text{WE}(A) - x, \end{cases}$$

and it follows that $\delta \in (N_{\Omega}(x))^0 = T_{\Omega}(x)$ and $\text{WE}(A) \subset x + H_{\delta}^+$. Since $A \subset \text{WE}(A)$ and $\overline{H_{\delta}^+} = \overline{P_{-\delta}}$, we deduce that

$$A \cap (x + \overline{P_{-\delta}}) = \emptyset \text{ for some } \delta \in T_{\Omega}(x)$$

and thus $x \notin \text{WE}(A, \Omega)$ according to Corollary 3.1. $\square$

Corollary 4.2. *Let X be a Hilbert space, $A \subset X$ be a nonempty compact set and $\Omega \subset X$ be a nonempty closed convex set. Then*

$$\text{SE}(A, \Omega) = \text{E}(A, \Omega) = \text{WE}(A, \Omega) = \text{Proj}_{\Omega}^{\gamma}(\text{co}(A)).$$

The following results obtained with an arbitrary norm will be useful in Section 4.3.

Proposition 4.3. *Let $A \subset X$ be a nonempty compact set, $\Omega \subset X$ be a nonempty closed convex set and γ be an arbitrary norm on X . Then we have*

$$\text{Proj}_{\Omega}^{\gamma}(A) \subset \text{WE}(A, \Omega).$$

Proof. Let $x \in \text{Proj}_{\Omega}^{\gamma}(A)$. Then $x \in \text{Proj}_{\Omega}^{\gamma}(y)$ for some $y \in A$ and by Proposition 4.1 there exists $p \in \partial\gamma(x - y)$ such that $-p \in N_{\Omega}(x)$. For any $\delta \in F_{\Omega}(x)$ and $\lambda > 0$ we deduce

$$\begin{aligned} \gamma(y - x - \lambda\delta) &\geq \langle -p, y - x - \lambda\delta \rangle \\ &= \langle p, x - y \rangle + \lambda \langle p, \delta \rangle \\ &= \gamma(x - y) + \lambda \langle p, \delta \rangle \\ &\geq \gamma(x - y). \end{aligned}$$

Consequently, $y - x \in \overline{P_{-\delta}}$, hence

$$A \cap (x + \overline{P_{-\delta}}) \neq \emptyset.$$

Since this condition is true for any $\delta \in F_{\Omega}(x)$, Proposition 3.2 shows that $x \in \text{WE}(A, \Omega)$. $\square$

Corollary 4.2. *Let $A \subset X$ be a nonempty compact set, $\Omega \subset X$ be a nonempty closed convex set and γ be an arbitrary norm on X . Then we have*

$$\text{Proj}_{\Omega}^{\gamma}(\text{WE}(A)) \subset \text{WE}(\text{WE}(A), \Omega).$$

Proof. Observe that $\text{WE}(A)$ is compact and substitute A by $\text{WE}(A)$ in the last proposition. $\square$

4.3. Projection properties in dimension 2

We will show now that, in dimension 2 and whatever the norm is, the *projection property* always holds for weakly efficient points but generally fails for efficient and strictly efficient points.

Proposition 4.5. *Assume that X is two dimensional and let $A \subset X$ be a nonempty compact set and $\Omega \subset X$ be a nonempty closed convex set. Then*

$$\text{WE}(A, \Omega) = \text{WE}(\text{WE}(A), \Omega).$$

Proof. The set A being contained in $\text{WE}(A)$ the inclusion

$$\text{WE}(A, \Omega) \subset \text{WE}(\text{WE}(A), \Omega)$$

is a direct consequence of Proposition 2.1.

Conversely, let $x \in \text{WE}(\text{WE}(A), \Omega)$. By Proposition 3.2 we get

$$\forall \delta \in F_{\Omega}(x), \exists \overline{a_{\delta}} \in \text{WE}(A) \text{ such that } \overline{a_{\delta}} \in x + \overline{P_{-\delta}}.$$

Since X is two dimensional, and according to the geometrical description of the Q_{δ} sets given in Section 3, we can assume without loss of generality that its closed complement $\overline{P_{-\delta}}$ contains a

closed half plane of type $J(u)$, $u \in \text{Ext}(B)$, such that $\overline{a\delta} \in x + J(u)$. Since $\overline{a\delta} \in \text{WE}(A)$ from Corollary 3.2 there exists $a \in A$ such that $a \in \overline{a\delta} + J(u)$. Hence $a \in \overline{a\delta} + J(u) \subset x + J(u) + J(u) = x + J(u) \subset x + \overline{P_{-\delta}}$. Then, $A \cap (x + \overline{P_{-\delta}}) \neq \emptyset$ for any $\delta \in F_{\Omega}(x)$ and we conclude by Proposition 3.2 that $x \in \text{WE}(A, \Omega)$. $\square$

To prove the *projection property* of weakly efficient points we need two technical lemmas.

Lemma 4.1. *Let $D \in \mathbb{R}^2$ be a connected set contained in an open half plane H^+ limited by a line passing through the origin. Then the cone generated by D and defined by $\text{cone}(D) := \{\lambda x; x \in D, \lambda > 0\}$ is convex and contained in H^+ .*

Proof. Any half plane the boundary of which is a line which contains the origin is a cone, thus we immediately obtain the inclusion $\text{cone}(D) \subset H^+$. Now suppose that $\text{cone}(D)$ is nonconvex. Then, there exist $x, y \in \text{cone}(D)$ and $\theta \in]0, 1[$ such that $z = \theta x + (1 - \theta)y \notin \text{cone}(D)$ and $z \neq 0$. Denoting by $H^+(z)$ (resp. $H^-(z)$) the open half plane passing through the origin and z and containing x (resp. containing y) and putting $O^+ = H^+(z) \cap H^+$ and $O^- = H^-(z) \cap H^+$, we see that O^+ and O^- are two nonempty disjoint open sets such that $D = (D \cap O^+) \cup (D \cap O^-)$ which contradicts the connectedness of D . $\square$

The result of Lemma 4.2 is valid in the general setting involving a compact set A and (instead of $\partial\gamma$) a multivalued mapping which is closed and has connected and compact values [19,20]. For simplicity we give an adapted proof of the following.

Lemma 4.2. *Let γ be a norm (a gauge) on X and let $A \subset X$ be a connected set. Then, $\bigcup_{a \in A} \partial\gamma(a)$ is connected.*

Proof. If $K := \bigcup_{a \in A} \partial\gamma(a)$ is not connected, there exist two closed sets F_1 and F_2 such that $F_1 \cap K \neq \emptyset$, $F_2 \cap K \neq \emptyset$, $K \subset F_1 \cup F_2$ and $K \cap F_1 \cap F_2 = \emptyset$. Then, consider the two nonempty subsets of A defined by $\hat{F}_i := \{x \in A, \partial\gamma(x) \cap F_i \neq \emptyset\}$, $i = 1, 2$, and let us show that $A = \hat{F}_1 \cup \hat{F}_2$, both $\hat{F}_1$ and $\hat{F}_2$ are nonempty, closed and disjoint,

contradicting A 's connectedness. The equality $A = \hat{F}_1 \cup \hat{F}_2$ directly follows from the definition of the $\hat{F}_i$'s. Each F_i intersection K , it intersects $\partial\gamma(x)$ for some $x \in A$ and thus each $\hat{F}_i$ is nonempty. To prove the closedness of $\hat{F}_i$, remark that any sequence (x_n) of elements of $\hat{F}_i$ converging to some x_∞ induces a sequence (p_n) such that $p_n \in \partial\gamma(x_n) \cap F_i$. The sequence (p_n) being contained in B^0 is bounded and thus we can assume without loss of generality that (p_n) converges to some $p_\infty \in F_i$. As the multivalued map $x \rightarrow \partial\gamma(x)$ is closed, we deduce that $p_\infty \in \partial\gamma(x_\infty)$. Hence $x_\infty \in \hat{F}_i$ and the closedness of $\hat{F}_i$ follows. Now, if $x \in \hat{F}_i$, we have $\partial\gamma(x) \cap F_1 \cap F_2 = \emptyset$ and as $\partial\gamma(x)$ is connected and is covered by $F_1 \cup F_2$, it cannot intersect both F_1 and F_2 and thus is included either in F_1 or in F_2 , hence $\hat{F}_1 \cap \hat{F}_2 = \emptyset$. $\square$

Lemmas 4.1 and 4.2 allow us to show the following result.

Proposition 4.6. *Assume that X is two dimensional. Let $A \subset X$ be a nonempty compact set, $\Omega \subset X$ be a nonempty closed convex set and γ be an arbitrary norm on X . Then*

$$\text{Proj}_{\Omega}^{\gamma}(\text{WE}(A)) = \text{WE}(\text{WE}(A), \Omega).$$

Proof. According to Corollary 4.4 we only have to prove the reverse inclusion $\text{WE}(\text{WE}(A), \Omega) \subset \text{Proj}_{\Omega}^{\gamma}(\text{WE}(A))$. However, to stress that $\text{Proj}_{\Omega}^{\gamma}(\text{WE}(A))$ and $\text{WE}(\text{WE}(A), \Omega)$ admit very close characterizations in terms of subdifferentials of the norm, we first give a similar but slightly different proof of the first inclusion.

From Proposition 4.1 we easily deduce that

$$x \in \text{Proj}_{\Omega}^{\gamma}(\text{WE}(A)) \iff -N_{\Omega}(x) \cap \left(\bigcup_{y \in \text{WE}(A)} \partial\gamma(x - y) \right) \neq \emptyset.$$

Then, substituting A by $\text{WE}(A)$ in the equivalence (see Proposition 2.2)

$$x \in \text{WE}(A, \Omega) \iff -N_{\Omega}(x) \cap \text{co} \left(\bigcup_{y \in A} \partial\gamma(x - y) \right) \neq \emptyset$$

we obtain

$$x \in \text{WE}(\text{WE}(A), \Omega) \iff -N_{\Omega}(x) \cap \text{co} \left(\bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y) \right) \neq \emptyset$$

and the inclusion $\text{Proj}_{\Omega}^{\gamma}(\text{WE}(A)) \subset \text{WE}(\text{WE}(A), \Omega)$ follows.

Let us now prove the reverse inclusion, and let $x \notin \text{Proj}_{\Omega}^{\gamma}(\text{WE}(A))$. By Proposition 4.1 we have

$$-N_{\Omega}(x) \cap \left(\bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y) \right) = \emptyset$$

and, since $N_{\Omega}(x)$ contains the origin, we get

$$0 \notin \bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y),$$

or equivalently according to Proposition 3.3

$$\exists \delta \neq 0, \bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y) \subset H_{\delta}^{+}.$$

As already noted the set $\text{WE}(A)$ is convex. Thus $x - \text{WE}(A)$ is convex, hence connected. The set

$$\bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y) = \bigcup_{z \in x - \text{WE}(A)} \partial \gamma(z)$$

being contained in the open half plane H_{δ}^{+} and connected by Lemma 4.2, Lemma 4.1 implies that cone $(\bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y))$ is convex and therefore contains $\text{co}(\bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y))$. Since $N_{\Omega}(x)$ is a cone, our assumption implies that

$$-N_{\Omega}(x) \cap \text{cone} \left(\bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y) \right) = \emptyset,$$

we get

$$-N_{\Omega}(x) \cap \text{co} \left(\bigcup_{y \in \text{WE}(A)} \partial \gamma(x - y) \right) = \emptyset$$

and the reverse inclusion is proved. $\square$

The following projection property of weakly efficient points and last result is a straightforward consequence of Propositions 4.5 and 4.6.

Theorem 4.2. *Let $A \subset X$ be a nonempty compact set, $\Omega \subset X$ be a nonempty closed convex set, γ be an arbitrary norm on X and assume that X is two dimensional. Then,*

$$\text{WE}(A, \Omega) = \text{Proj}_{\Omega}^{\gamma}(\text{WE}(A)).$$

In general, the *projection property* fails for efficient and strictly efficient points when the norm is not strictly convex. Consider again the example given in Section 2. With $A = \{a_1, a_2\}$ and $\Omega = [b_1, b_2]$, using Proposition 3.4 we get $\text{SE}(A) = A$, $\text{SE}(A, \Omega) = [b_1, c_1]$ and $\text{Proj}_{\Omega}^{\gamma}(\text{SE}(A)) = [c_1, c_2] \cup \{b_1\}$. With $A = \{a_2, a_4\}$ and $\Omega = [b_1, b_2]$, we have $E(A) = \text{co}\{a_2, c_2, a_4, c_3\}$, $\text{Proj}_{\Omega}^{\gamma}(E(A)) = [c_1, b_2]$ and $E(A, \Omega) = \{c_2\}$. This provides a case with a strict inclusion between $\text{Proj}_{\Omega}^{\gamma}(E(A))$ and $E(A, \Omega)$.

5. Conclusion

We have proved in this paper that weakly efficient points have the *projection property* in a Hilbert space and in any two dimensional normed space and we have shown that this property fails in general for efficient and strictly efficient points. In dimension $n \geq 3$ the projection property probably also fails for weakly efficient points as soon as some Q_{δ} sets are nonconvex. If we assume that all the Q_{δ} sets are convex the norm necessarily derives from an inner product [5] and thus the projection property holds in virtue of Theorem 4.1. However, we feel that some localization properties of efficient points and strictly efficient points could be obtained in dimension 2. In particular we conjecture that the inclusion $\text{SE}(A, \Omega) \subset \text{Proj}_{\Omega}^{\gamma}(\text{SE}(A))$ holds, at least in the polyhedral norm case, if $\text{SE}(A) = E(A)$ (i.e. if $\text{SE}(A)$ is connected). These localization properties will be investigated in a forthcoming paper.

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References

- [1] E. Carrizosa, E. Conde, F.R. Fernandez, J. Puerto, Efficiency in spatial location problems with non convex constraint, *Journal of Global Optimization* 6 (1995) 77–86.
- [2] E. Carrizosa, E. Conde, F.R. Fernandez, J. Puerto, Efficiency in Euclidean constrained location problems, *Operations Research Letters* 14 (1993) 291–295.
- [3] E. Carrizosa, F. Plastria, A characterization of efficient points in constrained location with regional demand, *Operations Research Letters* 19 (1996) 129–134.
- [4] E. Carrizosa, F. Plastria, Geometrical characterization of weakly efficient points, *Journal of Optimization Theory and Applications* 90 (1996) 217–223.
- [5] R. Durier, Sets of efficiency in a normed space and inner product, in: J. Jahn, W. Krabs (Eds.), *Lecture Notes in Econom. and Math. Systems, Recent Advances and Historical Development of Vector Optimization*, 1987, pp. 114–128.
- [6] R. Durier, On Pareto optima, the Fermat–Weber problem and polyhedral gauges, *Mathematical Programming* 47 (1990) 65–79.
- [7] R. Durier, C. Michelot, Set of efficient points in a normed space, *Journal of Mathematical Analysis and Applications* 117 (1986) 506–528.
- [8] B. Pelegrin, F.R. Fernandez, Determination of efficient points in multiple-objective location problems, *Naval Research Logistics* 35 (1988) 697–705.
- [9] F. Plastria, Continuous location problems and cutting plane algorithms, Thesis, Vrije Universiteit Brussel, 1983.
- [10] R.E. Wendell, A.P. Hurter, Location theory, dominance and convexity, *Operations Research* 21 (1973) 314–321.
- [11] R.E. Wendell, A.P. Hurter, T.J. Lowe, Efficient points in location problems, *AIIE Transactions* 9 (1977) 238–246.
- [12] R. Durier, C. Michelot, Geometrical properties of the Fermat–Weber problem, *European Journal of Operational Research* 20 (1985) 332–342.
- [13] R. Durier, C. Michelot, On the set of optimal points to the Weber problem: further results, *Transportation Science* 28 (1994) 141–149.
- [14] C. Michelot, Localization in multifacility location theory, *European Journal of Operational Research* 31 (1987) 117–184.
- [15] E. Carrizosa, E. Conde, M. Munoz, J. Puerto, Simpson points in planar problems with locational constraints. The round norm case, *Mathematics of Operations Research* 22 (1997) 276–290.
- [16] E. Carrizosa, E. Conde, M. Munoz, J. Puerto, Simpson points in planar problems with locational constraints. The polyhedral norm case, *Mathematics of Operations Research* 22 (1997) 291–300.
- [17] R.T. Rockafellar, *Convex Analysis*, Princeton Univ. Press, Princeton, NJ, 1970.
- [18] A.L. Brown, Best n -dimensional approximation to sets of functions, *Proceedings of the London Mathematical Society*, vol. 14, 1964, pp. 577–594.
- [19] E. Klein, A. Thompson, *Theory of Correspondences*, Canadian Mathematical Society, Wiley/Interscience, New York, 1984.
- [20] A.R. Waburton, Quasiconcave vector maximization: connectedness of the sets of Pareto-optimal and weak Pareto-optimal alternatives, *Journal of Optimization Theory and Applications* 40 (1983) 537–557.